FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Main)-2020

TEST DATE: 23-08-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. Using Gauss's Law

$$E4\pi r^2 = \frac{1}{\varepsilon_0}\int_0^r \frac{\alpha}{r}4\pi r^2 dr$$

$$E4\pi r^2 = \frac{4\pi\alpha}{\varepsilon_0}\frac{r^2}{2} \Rightarrow E = \frac{\alpha}{2\varepsilon_0} = \text{constant}$$

2. C

Sol. Acceleration of pulley, $a_1 = 2\ m/s^2$

Acceleration of the block, $a_2 = a_1 + 2\alpha R$

$a_2 = a_1 + 2a_1$ $[\because \alpha R = a_1]$

$a_2 = 3a_1$

$\Rightarrow a_2 = 3 \times 2 = 6\ m/s^2$

3. D

Sol. Time of flight, $T = \frac{2u\sin\theta}{2g} = \frac{u\sin\theta}{g}$

Horizontal range,

$$R = u\cos\theta\left(\frac{u\sin\theta}{g}\right) + \frac{1}{2}(2g)\left(\frac{u^2\sin^2\theta}{g^2}\right)$$

$$R = \frac{u^2\sin\theta\cos\theta}{g} + \frac{u^2\sin^2\theta}{g} = \frac{u^2}{g}\sin\theta(\cos\theta + \sin\theta)$$

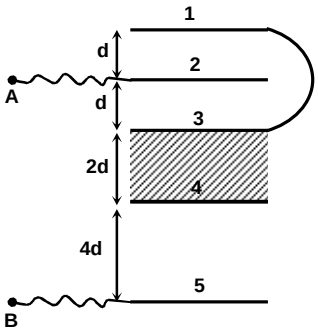


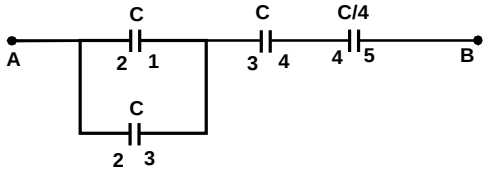
4. A

Sol. $C = \frac{\varepsilon_0 S}{d} = 1\mu F$

$\frac{1}{C_{AB}} = \frac{1}{2} + 1 + \frac{4}{1} = \frac{11}{2}$

$\therefore\ C_{AB} = \frac{2}{11}\mu F$

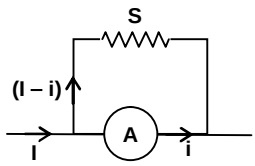




5. B

Sol. $(I - i)S = ir$

$S = \frac{ir}{(I - i)}$

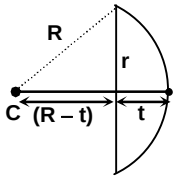


6. D

Sol. $E = -\frac{dV}{dr} = -\frac{ze}{4\pi \in_0}\left(-\frac{1}{r^2} + \frac{2r}{2R^3}\right) = \frac{ze}{4\pi\varepsilon_0}\left(\frac{1}{r^2} - \frac{r}{R^3}\right)$

7. C

Sol. Refractive index of the lens material is



$\mu = \frac{3\times10^8}{2\times10^8} = \frac{3}{2}$

Now, $R^2 = r^2 + (R - t)^2$

$R^2 = r^2 + R^2 - 2Rt + t^2$

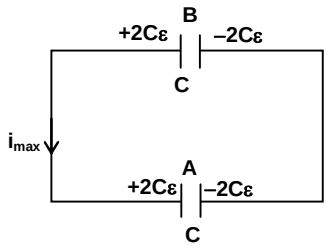
$R = \frac{r^2}{2t} = \frac{3\times3}{2\times0.3} = 15$ cm (as t^2 is very small, it can be neglected)

$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \left(\frac{3}{2} - 1\right)\left(\frac{1}{15}\right)$

f = 30 cm

8. A

Sol. When the current through the inductor is maximum. The two capacitors will be in parallel.



Now, $\frac{(3C\varepsilon)^2}{2C} + \frac{(C\varepsilon)^2}{2C} = 2\times\frac{(2C\varepsilon)^2}{2C} + \frac{1}{2}Li_{max}^2$

$\Rightarrow \frac{9C\varepsilon^2}{2} + \frac{C\varepsilon^2}{2} = 4C\varepsilon^2 + \frac{1}{2}Li_{max}^2$

$\Rightarrow\ i_{max} = \varepsilon\sqrt{\frac{2C}{L}}$

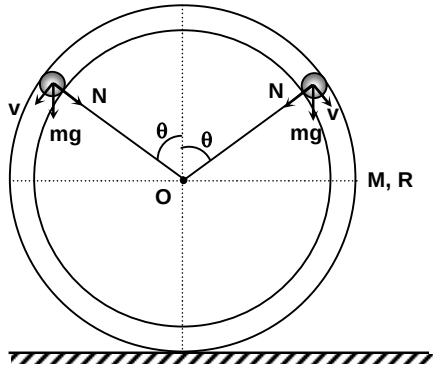
9. B

Sol. Since the tube loses contact with the horizontal surface when $\theta = 60°$

$2N\cos\theta = Mg$

$2N\cos 60° = Mg$

$N = Mg$ …(i)



Now, $mg\cos\theta + N = \frac{mv^2}{R}$

$\frac{mg}{2} + Mg = \frac{mv^2}{R}$ …(ii)

Using conservation of energy

$mgR(1-\cos\theta) = \frac{1}{2}m(v^2 - u^2)$

$\frac{mgR}{2} = \frac{m}{2}(v^2 - gR)$

$gR = v^2 - gR$

$v = \sqrt{2gR}$ ….(iii)

From (ii) and (iii)

$\frac{mg}{2} + Mg = 2mg$

$Mg = \frac{3mg}{2}$

$M = \frac{3m}{2} = 1.5$ kg

M = 1.5 kg

10. D

Sol. $\frac{W_1}{W_2} = \frac{qV_1}{qV_2} = \frac{\frac{KQ}{R}}{\frac{3}{2}\frac{KQ}{R}} = \frac{2}{3}$

11. B

Sol. Energy of each photon, $E = \frac{hc}{\lambda} = \frac{1240}{200} = 6.2$ eV

Now, $K_{max} = E - \phi = 6.2 - 3.8 = 2.4$ eV

Also, $eV_S = K_{max} = 2.4$ eV

$V_S = 2.4$ volt

12. C

Sol. Torque experienced by the loop is

$\tau = MB = \frac{\sqrt{3}}{4}\ell^2 IB$

Now, $\int \tau dt = \frac{5m\ell^2}{4}\omega$

$\frac{\sqrt{3}}{4}B\ell^2\int Idt = \frac{5m\ell^2}{4}\omega$

$\sqrt{3}BQ = 5m\omega$

$\omega = \frac{\sqrt{3}BQ}{5m}$

13. C

Sol. $\eta = \frac{E_2 I_2}{E_1 I_1} \times 100 = 83.3\%$

14. C

Sol. Fundamental frequency of a sonometer wire, $f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}}$

when tension is increased by 44%, then

$\Rightarrow \frac{f_1}{f} = \sqrt{\frac{1.44T}{T}} = 1.2$

$\therefore f_1 = 1.2f \Rightarrow f + 6 = 1.2f$

$\Rightarrow f = 30$ Hz

when the length is increased by 20%, then

$\frac{f_2}{f} = \frac{\ell}{\ell_2} = \frac{\ell}{1.2\ell}$

$f_2 = \frac{f}{1.2} = \frac{30}{1.2} = 25$ Hz

$\therefore$ decrease in the fundamental frequency = $f - f_2 = 5$ Hz

15. A

Sol. $t(\mu - 1) = 4\lambda$

$t = \frac{4\lambda}{(\mu - 1)} = 4.8\ \mu m$

16. A

Sol. To avoid disintegration

$\left(\frac{GM}{R^3}\right) r \geq \omega^2 r$

$\omega^2 \leq \frac{4\pi G\rho}{3}$

$\omega \leq \sqrt{\frac{4\pi G\rho}{3}}$

$\frac{2\pi}{T} \leq \sqrt{\frac{4\pi G\rho}{3}}$

$T \geq \sqrt{\frac{3\pi}{G\rho}}$

$T \geq \sqrt{\frac{3\pi}{\frac{20}{3}\times 10^{-11} \times \frac{9\pi}{20}\times 10^{11}}}$

$T \geq 1$ sec

17. C

Sol. Electric field 10 V/cm

Distance moved 1 cm

Time taken 20 μsec

$\therefore$ velocity of minority carrier = $\frac{1\,cm}{20\times 10^{-6}\,sec} = 50{,}000$ cm/s

Drift velocity $v_d = \mu E$

$\mu = \frac{V_d}{E} = \frac{50,000}{10} = 5000\ cm^2/V\text{-}s$

18. B

Sol. $H = i^2Rt = ms\Delta\theta$

$(0.1)^2\ 10^3 t = 10 \times 10^3 \times 4.2 \times 10$

$t = \frac{420}{0.01} = 42000\ sec$

$t = 700$ minutes

19. C

Sol. Since diode is forward biased so it behaves as short circuit.

Equivalent resistance of circuit = 15 kΩ

Current through the cell = $\frac{30}{15} = 2\ mA$

Current between A and B = 1 mA

$V_{AB} = 10\ k\Omega \times 1\ mA = 10\ V$

20. B

Sol. $f_0 = 1.5$ cm, $f_e = 5$ cm

$\frac{1}{v} - \frac{1}{-2} = \frac{1}{1.5}$

$\frac{1}{v} = \frac{1}{1.5} - \frac{1}{2}$

$v = +6$ cm

Hence the magnification for the adjustment for near point vision is

$m = \frac{v}{u}\left(1 + \frac{D}{f_e}\right) = \frac{-6}{2}\left(1 + \frac{25}{5}\right) = -18$

$|m| = 18$

SECTION – B

21. 5

Sol. $F_1 = \rho\left(\pi R^3 - \frac{2}{3}\pi R^3\right)g = \frac{\rho\pi R^3}{3}g$, $F_2 = \rho\left(\pi R^3 + \frac{2}{3}\pi R^3\right)g = \frac{5\rho\pi R^3}{3}g$

$\therefore F_2/F_1 = 5$

22. 4

Sol. $m\frac{d^2x}{dt^2} = -16kx$

$\frac{d^2x}{dt^2} = -\left(\frac{16k}{m}\right)x$

$\therefore$ frequency of small oscillations $f = \frac{1}{2\pi}\sqrt{\frac{16k}{m}} = \frac{4}{2\pi}\sqrt{\frac{k}{m}}$

$\therefore n = 4$

SECTION – C

23. 00600.00

Sol. With both switches open: $I_A = \dfrac{E}{R_1 + R_2 + R_3}$

With both switches closed:
Current through E and R_2

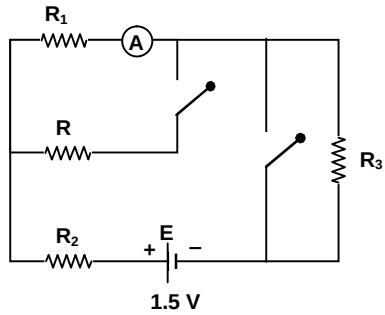
$$I' = \frac{E}{R_2 + \dfrac{RR_1}{R + R_1}}$$

Current through the ammeter

$$I'_A = \frac{RI'}{R + R_1}$$

$$I_A = I'_A$$

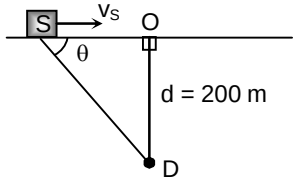
$$R = \frac{R_1R_2}{R_3} = 600\Omega$$



24. 00625.00

Sol. $\cos\theta = \dfrac{v_s t}{vt} = \dfrac{v_s}{v} = \dfrac{66}{330} = \dfrac{1}{5} = 0.2$

$$f = \left(\frac{v}{v - v_s\cos\theta}\right) f_0 = \frac{f_0}{\left(1 - \dfrac{1}{25}\right)} = \frac{25}{24} \times 600 = 625\ \text{Hz}$$



25. 00015.38

Sol. Total work done by gas per cycle, $\Delta W_{cycle} = P_0V_0$

During the process AB

$$\Delta Q_{AB} = nC_V\Delta T = n\frac{3R}{2}\Delta T = \frac{3}{2}V_0(2P_0 - P_0) = \frac{3}{2}P_0V_0$$

During the process BC

$$\Delta Q_{BC} = nC_P\Delta T = n\frac{5R}{2}\Delta T = \frac{5}{2} \times 2P_0(2V_0 - V_0) = 5P_0V_0$$

Efficiency of the cycle

$$\eta = \frac{\Delta W_{cycle}}{\Delta Q_{supplied}} \times 100 = \frac{P_0V_0}{\left(\dfrac{3P_0V_0}{2} + 5P_0V_0\right)} \times 100 = \frac{2}{13} \times 100 = 15.38\%$$

Chemistry

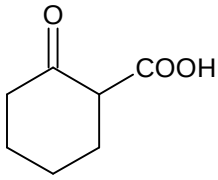
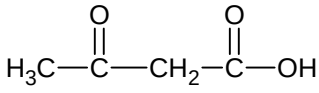
PART – II

SECTION – A

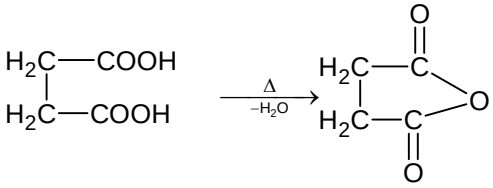
26. C

Sol. Cassiterite $\rightarrow SnO_2$
Magnetite $\rightarrow Fe_3O_4$
Pyrolusite $\rightarrow MnO_2$
Malachite $\rightarrow CuCO_3.Cu(OH)_2$

27. D

Sol.  and 

are β-keto acid which undergo decarboxylation on heating.
β-γ unsaturated acid also decarboxylates on heating.



28. D

Sol. Let mass of solute = x g, mass of water = (212 – x) g

Then $\dfrac{x/50}{x/50+\left(\dfrac{212-x}{18}\right)} = 0.1 \Rightarrow x = 50\,g$

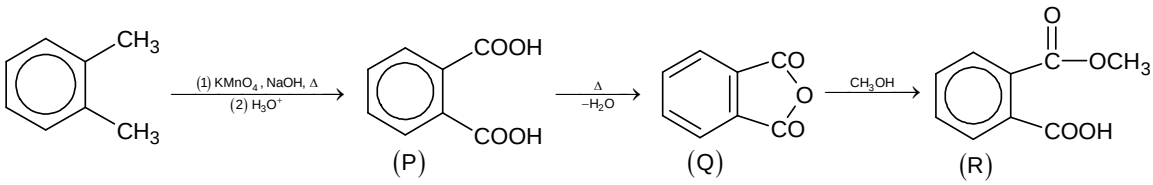
In solution left over, mass of the solute = 50 – 14 = 36.0 g
Mass of water = 180 – 36.0 = 144 g

Now molality (left over solution) $m = \dfrac{36}{50} \times \dfrac{1000}{144} = 5.0$

29. C

Sol. Peroxyacyl nitrate $\left(R-\overset{\overset{O}{\|}}{C}-O-O-NO_2 \right)$, Acrolein $CH_2 = CH - CHO$ and ozone are major components of photochemical smog.

30. D

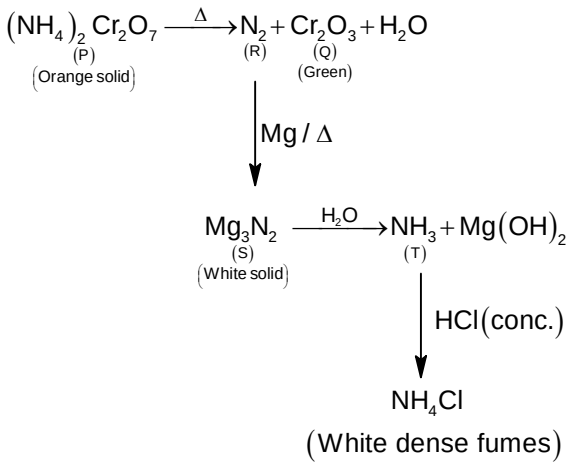
Sol. 

31. A

Sol. The viscosity of hydrophilic sol is much higher and surface tension is much lower than that of water.

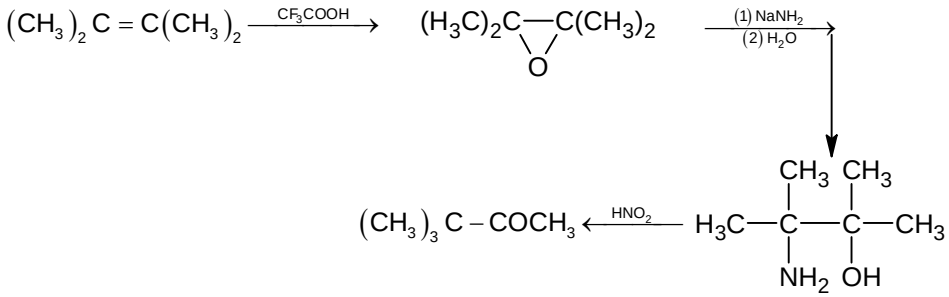
32. A

Sol.



33. C

Sol.



34. D

Sol. $P_A V = \frac{3}{M_A} RT$ and $P_B V = \frac{9}{M_B} .RT$

But $P_B = 2.0 - 1.5 = 0.5$ atm

So, $\frac{P_A}{P_B} = \frac{1.5}{0.5} = \frac{3M_B}{9M_A}$

$\Rightarrow \frac{M_B}{M_A} = 9$

$\frac{V_A}{V_B} = \sqrt{\frac{M_B}{M_A}} = 3.$

35. B

Sol. Ortho-hydrogen has proton of parallel spin and magnetic moment of para form is zero, while ortho has twice that of proton.

Percentage of para-hydrogen increases with decrease in temperature and at room temperature ortho-hydrogen is more stable than para-hydrogen.

36. B

Sol. pH (HA solution) $= pK_a + \log\frac{[A^-]}{[HA]}$

$= -\log 1.0\times10^{-5} + \log\frac{0.15}{0.3-0.15}$

= 5.0

pH = 5.0 remains unaffected on dilution [Buffer solution].

For half neutralization of 'V' volume of the acids, V/2 volume of NaOH would be needed.

Hence, molarity of HCl solution after half-neutralisation

$= \frac{0.3\times V - 0.3\times V/2}{V + \frac{V}{2}} = \frac{(0.3-0.15)V}{1.5V} = 0.1$

$= [H^+]$

On 10 times dilution $[H^+] = \frac{0.1}{10} = 0.01 = 1\times10^{-2}$

pH = 2.

37. C

Sol. Setting of plaster of paris is retarded by addtition of borax.

38. C

39. B

Sol. Bell metal $\rightarrow$ Cu and Sn

Type metal $\rightarrow$ Pb, Sn, Sb

Monel metal $\rightarrow$ Cu, Ni, Fe

Gun metal $\rightarrow$ Cu, Zn, Sn

40. B

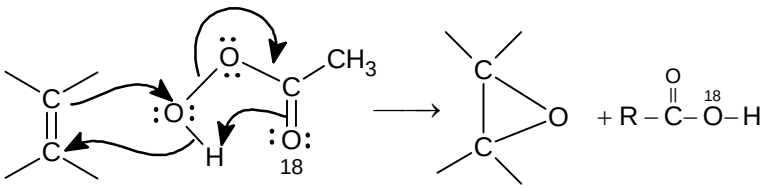
Sol. With Gilman's reagent [$(CH_3)_2CuLi$] the 1-4 addtion product is the exclusively major product.

41. B

Sol. B_2H_6, Al_2Cl_6, $Al_2(CH_3)_6$ are non-planar.

42. C

Sol.



43. C

Sol. Passing of H_2S gas through a solution of an oxidizing agent like HNO_3, $KMnO_4$, SO_2 form colloidal sulphur.

$2HNO_3 + H_2S \longrightarrow 2NO_2 + 2H_2O + \underset{(\text{Colloidal form})}{S}$

$SO_2 + 2H_2S \longrightarrow 2H_2O + \underset{(\text{Colloidal form})}{3S}$

44. B

Sol. Furfural test $\rightarrow$ Fructose gives purple (violet) colour on addition of alcoholic α-Naphthol and conc. HCl, whereas glucose gives no colour.

45. B

Sol. Heat capacity is an extensive property as it depends upon amount of substance in the sample. Dielectric constant, viscosity and standard electrode potential are intensive property.

SECTION – B

46. 6

Sol. Let equal masses be m g.

Then amount of 'X' left out $= 10 \times \frac{m}{10} \times \left(\frac{1}{2}\right)^{30/10}$

$= m\left(\frac{1}{2}\right)^3$

and amount of 'Y' left out $= 20 \times \frac{m}{20} \times \left(\frac{1}{2}\right)^{30/t}$

$= m\left(\frac{1}{2}\right)^{30/t}$

Then $\frac{\text{mass of 'X'}}{\text{mass of 'Y'}} = \frac{m \times \left(\frac{1}{2}\right)^3}{m \times \left(\frac{1}{2}\right)^{30/t}}$

$\frac{4}{1} = \frac{m \times \left(\frac{1}{2}\right)^3}{m \times \left(\frac{1}{2}\right)^{30/t}} \Rightarrow \frac{1}{4} \times \left(\frac{1}{2}\right)^3 = \left(\frac{1}{2}\right)^{30/t}$

$\Rightarrow \left(\frac{1}{2}\right)^5 = \left(\frac{1}{2}\right)^{30/t}$

$\Rightarrow$ t = 6 days.

47. 6

Sol. Nylon-6,6 ; Glyptal ; Dacron ; Bakelite ; BUNA-S ; Polyurethrane are copolymers.

SECTION – C

48. 00014.14

Sol. Volume of N_2 at STP = $\frac{P_1V_1}{T_1} \times \frac{T_2}{P_2} = \frac{756}{300} \times 50 \times \frac{273}{760}$

= 45.26 ml

% of N_2 $\frac{28}{22400} \times \frac{45.26}{0.40} \times 100 = 14.14$

49. 00007.65

Sol. $\sqrt{\ell(\ell+1)}\frac{h}{2\pi} = \sqrt{3}\frac{h}{\pi}$

$\Rightarrow \ell = 3$, this means the electron belongs to n = 4(principal shell)

$E_n = -13.6\frac{z^2}{n^2} = -13.6 \times \frac{3^2}{4^2} = -7.65$ eV

50. 00000.17

Sol. $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$

Total number of moles at eq. $1 + \alpha$

$1-\alpha \qquad \alpha \qquad \alpha$

$$M = \frac{dRT}{P} = \frac{1.162 \times 0.082 \times 750}{1.0} = 71.46$$

$$71.46 = \frac{99}{1+\alpha} \Rightarrow \alpha = 0.385$$

$$K_P = \frac{\alpha^2}{1-\alpha^2} = \frac{(0.385)^2}{1-(0.385)^2} = 0.17$$

Mathematics

PART – III

SECTION – A

51. D

Sol. Since a, b, c are the sides of a triangle $\therefore$ a, b, c are real $\Rightarrow$ 1 + i is the other root

Sum of the roots = 2 and product = 2 $\Rightarrow$ equation is $x^2 - 2x + 2 = 0$

On comparing with $ax^2 - bx + c = 0$, we have a = k, b = 2k, c = 2k

Hence, triangle is isosceles greatest angle is B or C

$= \cos^{-1}\left(\frac{(2k)^2 + (k)^2 - (2k)^2}{2\cdot 2k\cdot k}\right) = \cos^{-1}\left(\frac{1}{4}\right)$

Least angle is $A = \cos^{-1}\left(\frac{(2k)^2 + (2k)^2 - k^2}{2\cdot 2k\cdot 2k}\right) = \cos^{-1}\left(\frac{7}{8}\right)$

Also, a = k, 2b = 4k, 3c = 6k are not in Arithmetic Progression

52. B

Sol. 1st term of the successive brackets are 2^0, 2^1, 2^{1+2}, 2^{1+2+3}, ...

$\therefore$ 1st term of 11th bracket is $2^{1+2+3+.....+10} = 2^{\frac{10\times 11}{2}} = 2^{55}$

Sum of all terms in 11th bracket is $2^{55} + 2^{56} + 2^{57} + ...$ to 11th terms $= \frac{2^{55}\left(2^{11}-1\right)}{2-1} = 2^{66} - 2^{55}$

$\therefore$ p : q = 66 : 55 = 6 : 5. Now, $\frac{2p+q}{2p-q} = \frac{2\left(\frac{p}{q}\right)+1}{2\left(\frac{p}{q}\right)-1} = \frac{2\times\frac{6}{5}+1}{2\times\frac{6}{5}-1} = \frac{17}{7}$

53. D

Sol. Let the number of sides be n, $\frac{{}^nC_4}{{}^nC_3} = \frac{2}{1}$

$\Rightarrow \frac{n(n-1)(n-2)(n-3)\times 3!}{4!\times n(n-1)(n-2)} = \frac{2}{1} \Rightarrow \frac{n-3}{4} = 2 \Rightarrow n = 11$

54. B

Sol. $|A| = \sqrt[3]{8\cdot 2\cdot\frac{1}{4}} = \sqrt[3]{4} = 4^{\frac{1}{3}}$ $\because$ $\left|\text{Adj}(\text{Adj } A)\right| = |A|^{(n-1)^2} = \left(4^{\frac{1}{3}}\right)^{(3-1)^2} = \left(4^{\frac{1}{3}}\right)^4 = \sqrt[3]{4^4} = \sqrt[3]{\lambda^\lambda} \Rightarrow \lambda = 4$

Hence, $2\lambda - 1 = 2\times 4 - 1 = 7$

55. D

Sol. $\because$ A and B occurs at the end in 2 ways (A ..... B or B ..... A) other 4 places can be filled by A, N, A, N in $\frac{4!}{2!\times 2!} = 6$ ways

$\therefore$ Number of ways in which A and B occurs at the ends = $2\times 6 = 12$

$\therefore$ Required probability = $\frac{12}{\frac{6!}{2!\times 2!}} = \frac{1}{15} \Rightarrow 3p = \frac{1}{5}$

(A) Perpendicular distance of $3x - 4y - 1 = 0$ from origin = $\frac{|-1|}{\sqrt{3^2+4^2}} = \frac{1}{5}$

(B) $e = \sqrt{1-\frac{b^2}{a^2}} = \sqrt{1-\frac{24}{25}} = \sqrt{\frac{1}{25}} = \frac{1}{5}$

(C) $\lim\limits_{x\to\infty} \frac{[x]-5\ln x}{5(x-\{x\})} = \lim\limits_{x\to\infty} \frac{x-5\ln x}{5x} = \frac{1}{5} - \lim\limits_{x\to\infty} \frac{\ln x}{x}$ ($\because$ [x] = x when $x \to \infty$) = $\frac{1}{5} - 0 = \frac{1}{5}$

(D) Distance between $3x - 4y + 5 = 0$ and $3x - 4y + 10 = 0$ is = $\frac{|10-5|}{\sqrt{3^2+4^2}} = \frac{5}{5} = 1$

Hence, (D) is the correct option.

56. B

Sol. Let $\cos\theta = x$, $\sqrt{2x^2-4x+6} + \sqrt{19-6x+3x^2} = 5 - x^2 + 2x$

or $\sqrt{2(x^2-2x+1)+4} + \sqrt{3(x^2-2x+1)+16} = 6 - (x^2-2x+1)$

or $\sqrt{2(x-1)^2+4} + \sqrt{3(x-1)^2+16} = 6 - (x-1)^2$

L.H.S. ≥ 6 and R.H.S. ≤ 6. But L.H.S. = R.H.S.

Hence, $(x-1)^2 = 0$, $x = 1$, $\cos\theta = 1$, $\theta = 2n\pi$, $\theta = -2\pi, 0, 2\pi$

Number of solution = 3

57. A

Sol. $\because$ $\operatorname{cosec}\theta = \cot\theta + 1$

$\therefore$ $\operatorname{cosec}\theta - \cot\theta = 1$ ..... (1)

But $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

$\Rightarrow$ $\operatorname{cosec}\theta + \cot\theta = 1$ ..... (2)

From equation (1) and (2), we get $\operatorname{cosec}\theta = 1 \Rightarrow \sin\theta = 1$

$x = 1 = \sin\theta$ is one of the roots of equation

58. C

Sol. $\cos^{-1}\left(\frac{xy}{2\cdot 3} - \sqrt{1-\frac{x^2}{4}}\cdot\sqrt{1-\frac{y^2}{9}}\right) = \theta \Rightarrow (xy - 6\cos\theta)^2 = (4-x^2)(9-y^2)$

$\Rightarrow 9x^2 + 4y^2 - 12xy\cos\theta = 36\sin^2\theta \Rightarrow P = 36\sin^2\theta$

(A) $\frac{1}{18}\int_0^{\pi} P\,d\theta = \int_0^{\pi} 2\sin^2\theta = \int_0^{\pi}(1-\cos 2\theta)d\theta = \left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi} = \pi$

(B) $I = \int_0^{\pi/2} \frac{6\sin\theta}{6\sin\theta + \sqrt{36-36\sin^2\theta}}d\theta = \int_0^{\pi/2} \frac{\sin\theta\,d\theta}{\sin\theta+\cos\theta}$ ..... (1)

$I = \int_0^{\pi/2} \frac{\cos\theta\,d\theta}{\cos\theta+\sin\theta}$ (Using property) ..... (1)

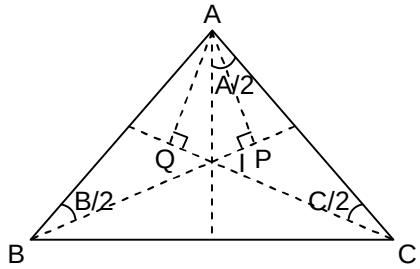
Adding equation (1) and (2), we get $2I = \int_0^{\pi/2} d\theta = \frac{\pi}{2}$, $I = \frac{\pi}{4}$

(C) $0 \leq 36\sin^2\theta \leq 1$ hence minimum value of P is zero

(D) Maximum value of P is 36. Hence, (C) is correct option

59. D

Sol. $\because \ \frac{AQ}{AC} = \sin\left(\frac{C}{2}\right) \Rightarrow AQ = AC\sin\left(\frac{C}{2}\right)$ ..... (1)



$$\frac{CI}{\sin\frac{A}{2}} = \frac{AC}{\sin\left(\pi-\left(\frac{A}{2}+\frac{C}{2}\right)\right)} \text{ in } \Delta AIC$$

$$\text{or } CI \equiv AC\cdot\frac{\sin\left(\frac{A}{2}\right)}{\sin\left(\frac{A+C}{2}\right)} = AC\frac{\sin\left(\frac{A}{2}\right)}{\cos\left(\frac{B}{2}\right)} \quad \text{..... (2)}$$

Dividing (1) by (2), we get $\frac{AQ}{CI} = \frac{\cos\frac{B}{2}\cdot\sin\frac{C}{2}}{\sin\frac{A}{2}}$. Similarly $\frac{AP}{BI} = \frac{\cos\frac{C}{2}\cdot\sin\frac{B}{2}}{\sin\frac{A}{2}}$

$$\frac{AQ}{CI}+\frac{AP}{BI} = \frac{\sin\left(\frac{B}{2}+\frac{C}{2}\right)}{\sin\frac{A}{2}} = \frac{\sin\left(\frac{B+C}{2}\right)}{\cos\left(\frac{B+C}{2}\right)} = \tan\left(\frac{B+C}{2}\right)$$

60. A

Sol. Let $g(x) = \tan^{-1}\left(\frac{x}{\sqrt{1+x^2}-1}\right)$, put x = tan θ

$$= \tan^{-1}\left(\frac{\tan\theta}{\sec\theta-1}\right) = \tan^{-1}\left(\frac{\sin\theta}{1-\cos\theta}\right) = \tan^{-1}\left(\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}}\right)$$

$$= \tan^{-1}\left(\cot\frac{\theta}{2}\right) = \tan^{-1}\left(\tan\left(\frac{\pi}{2}-\frac{\theta}{2}\right)\right) = \frac{\pi}{2}-\frac{\theta}{2} = \frac{\pi}{2}-\frac{1}{2}\tan^{-1}x$$

$$f(x) = g'(x) = -\frac{1}{2}\cdot\frac{1}{1+x^2}, \ f(x^2) = -\frac{1}{2(1+x^4)}$$

$$\therefore \ \frac{1}{f(x)}+\frac{1}{f(x^2)} = -2(1+x^2)-2(1+x^4) = -(2x^4+2x^2+4) = -2(x^4+x^2+2)$$

61. B

Sol. Let $P(x_1, y_1)$ on the curve $\frac{dy}{dx} = -\frac{x^3}{y^3} = -\frac{x_1^3}{y_1^3}$. Equation of tangent at $P(x_1, y_1)$

$$y-y_1 = -\frac{x_1^3}{y_1^3}(x-x_1), \ y_1^3\cdot y - y_1^4 = -x\cdot x_1^3 + x_1^4, \ y_1^3\cdot y + x\cdot x_1^3 = x_1^4+y_1^4 = c$$

$$\frac{y}{\frac{c}{y_1^3}}+\frac{x}{\frac{c}{x_1^3}} = 1, \ a = \frac{c}{x_1^3} \text{ and } b = \frac{c}{y_1^3}, \ x_1 = \left(\frac{c}{a}\right)^{\frac{1}{3}}, \ y_1 = \left(\frac{c}{b}\right)^{\frac{1}{3}}, \ x_1^4+y_1^4 = c, \ \left(\frac{c}{a}\right)^{\frac{4}{3}}+\left(\frac{c}{b}\right)^{\frac{4}{3}} = c$$

$$a^{-\frac{4}{3}}+b^{-\frac{4}{3}} = c^{-\frac{1}{3}}$$

$$\Rightarrow \alpha = \frac{4}{3}, \ \beta = \frac{2}{3}, \ \gamma = \frac{1}{3}$$

$\Rightarrow \alpha, \beta, \gamma$ are in Geometric Progression

62. B

Sol. $I = \int \frac{x^{-1} + x^{-3} + x^5 + ... + x^{-2021}}{1 + x^2 + x^4 + ... + x^{2020}} dx = \int \frac{\frac{1}{x} + \frac{1}{x^3} + \frac{1}{x^5} + ... + \frac{1}{x^{2021}}}{1 + x^2 + x^4 + ... + x^{2020}} dx$

$= \int \frac{\frac{x^{2020} + x^{2018} + x^{2016} + ... 1}{x^{2021}}}{1 + x^2 + x^4 + ... + x^{2020}} dx = \int \frac{x^{-2021}\left(1 + x^2 + x^4 + ... + x^{2016} + x^{2018} + x^{2020}\right)}{1 + x^2 + x^4 + ... + x^{2020}} dx$

$= \int x^{-2021} dx = \frac{x^{-2021+1}}{-2021+1} + c = c - \frac{1}{2020x^{2020}} + c$. Hence k = 2020

63. B

Sol. $I_2 - I_1 = \int_1^3 \left(3x^2 - 4x - 5\right) f\left(x^3 - 2x^2 - 5x + 2020\right) dx$. Put $x^3 - 2x^2 - 5x + 2020 = z$

$(3x^2 - 4x - 5)dx = dz$, when $x = 1$, $1 - 2 - 5 + 2020 = z_{min}$, $z_{min} = 2014$

When $x = 3$, $27 - 2 \times 9 - 5 \times 3 + 2020 = z_{max}$, $z_{max} = 2014 \Rightarrow I_2 - I_1 = \int_{2014}^{2014} f(z) dz = 0$

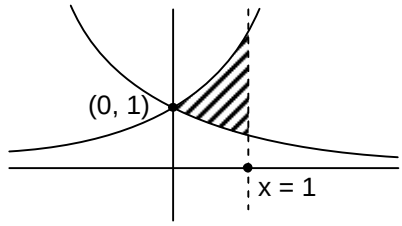
$I_2 = I_1 \Rightarrow \frac{2I_1}{3I_2} = \frac{2}{3} \rightarrow (0, 1)$

64. B

Sol. $y^2 - 4x - 4y + 12 = 0 \Rightarrow y^2 - 4y + 4 = 4x - 8$

$\Rightarrow (y - 2)^2 = 4(x - 2)$ directrix of the above curve is

Required area



$= \int_0^1 \left(e^x - e^{-x}\right) dx = \left[e^x + e^{-x}\right]_0^1 = \left(e + e^{-1}\right) - 2$

$= \left(1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} .....\right) + \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + ...\right) - 2$

$= 2\left(\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + ...\right) = 2\sum_{r=1}^{\infty} \frac{1}{(2r)!}$

65. D

Sol. $f''(x) = g''(x) \Rightarrow f'(x) = g'(x) + c \Rightarrow f'(2019) = g'(2019) + c \Rightarrow c = 0 \Rightarrow f'(x) = g'(x)$

$\Rightarrow f(x) = g(x) + d \Rightarrow f(2018) = g(2018) + d \Rightarrow 3 = 2 + d \Rightarrow d = 1 \Rightarrow f(x) = g(x) + 1$

$\Rightarrow f(2020) - g(2020) = 1$

66. A

Sol. $\sqrt{9 + (2x - y)^2} = 1 - (2x - y) \Rightarrow 9 + (2x - y)^2 = 1 + (2x - y)^2 - 2(2x - y)$

$\Rightarrow 9 = 1 - 4x + 2y \Rightarrow y = 2x + 4$

67. B

Sol. Consider

p	q	~p	~q	~p ∨ ~q	~q ∨ p	(p ∧ q)	~(p ∧ q)	p ∧ ~q	q ∨ p
T	T	F	F	F	T	T	F	F	T
T	F	F	T	T	T	F	T	T	T
F	T	T	F	T	F	F	T	F	T
F	F	T	T	T	T	F	T	F	F

Therefore ~(p ∧ q) = ~p ∨ ~q

68. B

Sol. Let e_1 and e_2 be the eccentricities respectively $e_1 = \frac{1}{e_2}$

$\therefore\ 9 = 16\left(1-e_2^2\right) \Rightarrow e_2^2 = \frac{7}{16} \Rightarrow e_1^2 = \frac{16}{7}$ also $b^2 = 7\left(e_1^2 - 1\right) = 7\left(\frac{16}{7} - 1\right) = 7 \times \left(\frac{16-7}{7}\right) = 9$

$\Rightarrow b = 3\ \therefore$ area bounded by the curve $\frac{x^2}{25} + \frac{y^2}{b^2} = 1$ is $\pi(5) \times 3 = 15\pi$

69. C

Sol. Putting z = 1, $\frac{x-6}{3} = \frac{y-2}{5} = \frac{z+3}{2} \Rightarrow \frac{x-6}{3} = \frac{y-2}{5} = \frac{1+3}{2} \Rightarrow \frac{x-6}{3} = \frac{y-2}{5} = 2$

$\Rightarrow x = 12, y = 12$. One putting $x = 12, y = 12$ in $xy = a^2$, we get $a^2 = 144 \Rightarrow a = \pm 12$
Two real value of a

70. C

Sol. $\vec{a} + \vec{b} = \vec{p},\ 2\vec{a} + \vec{b} = \vec{q} \Rightarrow \vec{a} = -3\hat{i} + 2\hat{j},\ \vec{b} = 5\hat{i} - 5\hat{j}$

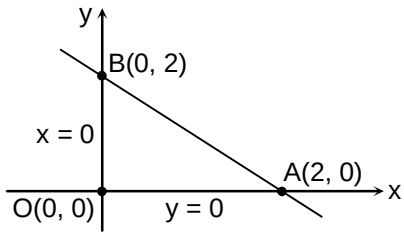
$\therefore\ \cos\theta = \frac{a \cdot b}{|\vec{a}||\vec{b}|} = \frac{-15-10}{\sqrt{13} \cdot \sqrt{50}} = \frac{-25}{\sqrt{13} \times 5\sqrt{2}} = -\sqrt{\frac{25}{26}}$

$\therefore\ \cos\theta + \sqrt{\frac{\alpha}{\alpha+1}} = 0 \Rightarrow -\sqrt{\frac{25}{26}} + \sqrt{\frac{\alpha}{\alpha+1}} = 0 \Rightarrow \alpha = 25$

SECTION – B

71. 1

Sol. $\because\ xy(2 - x - y) = 0$
$\Rightarrow$ 3 sides of the triangle are $x = 0$, $y = 0$ and $x + y = 2$
Clearly $H(\alpha, \beta)$ is at origin $\Rightarrow \alpha = 0, \beta = 0$
The triangle is right angled hence circumcentre
$C(\alpha_2, \beta_2) \equiv (1, 1) \Rightarrow \alpha_2 = \beta_2 = 1$.
Centroid $G(\alpha_1, \beta_1) = \left(\frac{2}{3}, \frac{2}{3}\right)$. Hence the value of



$5 \times 0 - 4 \times 0 + 3 \times \frac{2}{3} + 3 \times \frac{2}{3} - 2 \times 1 - 1 = 0 \Rightarrow 4 - 2 - 1 = 1$

72. 3

Sol. $A\left(10^{34} - 1\right) = \underbrace{272727.....27}_{34-\text{digits}}$ = 27 + 27 × 10^2 + 27 × 10^4 + ... 17 terms

$= \frac{27\left((10^2)^{17} - 1\right)}{10^2 - 1} = \frac{27}{99}\left(10^{34} - 1\right) = \frac{3}{11}\left(10^{34} - 1\right) \Rightarrow A = \frac{3}{11} \Rightarrow 11A = 3$

SECTION – C

73. 00002.87

Sol. We know that standard deviation of first n natural numbers = $\sqrt{\frac{n^2 - 1}{12}}$

Put n = 10, we get standard deviation = $\sqrt{\frac{99}{12}} = \sqrt{8.25} = 2.87$

74. 00001.41

Sol. Let $P \equiv \left(n, \sqrt{n^2+1}\right)$, $d_n = \dfrac{\sqrt{n^2+1}-n}{\sqrt{2}}$ or $4n \cdot d_n = \dfrac{4n \cdot \left(\sqrt{n^2+1}-n\right)}{\sqrt{2}} = 2\sqrt{2}\,n\left(\sqrt{n^2+1}-n\right)$

$$\lim_{n\to\infty}\left(4n \cdot d_n\right) = \lim_{n\to\infty} 2\sqrt{2}\,n\left(\sqrt{n^2+1}-n\right) = \lim_{n\to\infty} \frac{2\sqrt{2}\,n\left(n^2+1-n^2\right)}{\sqrt{n^2+1}+n}$$

$$\lim_{n\to\infty} \frac{2\sqrt{2}\,n}{n\left(\sqrt{1+\dfrac{1}{n^2}}+1\right)} = \frac{2\sqrt{2}}{2} = \sqrt{2} = 1.414$$

75. 00018.66

Sol. 6, a, b are in Harmonic Progression $\Rightarrow \dfrac{1}{6}, \dfrac{1}{a}, \dfrac{1}{b}$ are in Arithmetic Progression

$$\Rightarrow \frac{2}{a} = \frac{1}{6} + \frac{1}{b} \Rightarrow \frac{1}{b} = \frac{2}{a} - \frac{1}{6} \Rightarrow b = \frac{6a}{12-a} \Rightarrow a \in \{3, 4, 6, 8, 9, 10, 11\}$$

Hence, the value of N is 7. So the value of $\dfrac{8N}{3} = \dfrac{8\times 7}{3} = \dfrac{56}{3} = 18.6666$